



NEPSE Stock Market Prediction and Analysis

Introduction

The **Nepal stock market** generates large amounts of **trading data** every day. Investors face difficulty understanding **stock movements** and identifying important factors affecting **stock prices**.

Problem Statement

Predicting **stock price movement in NEPSE** is difficult because market performance depends on multiple variables such as **trading volume, company sector, historical prices,** and **market trends**.

This project aims to analyze **stock data** and build **predictive models** for **stock movement**.

Objectives

Main Objective

To analyze **historical NEPSE** stock market data and **predict stock price movement** using **statistical** and **predictive modeling** techniques.

Specific Objectives

- Analyze historical NEPSE stock trends.
- Identify factors affecting stock price movement.
- Apply predictive modeling techniques.
- Evaluate prediction performance and accuracy.

Dataset Selection

Dataset:

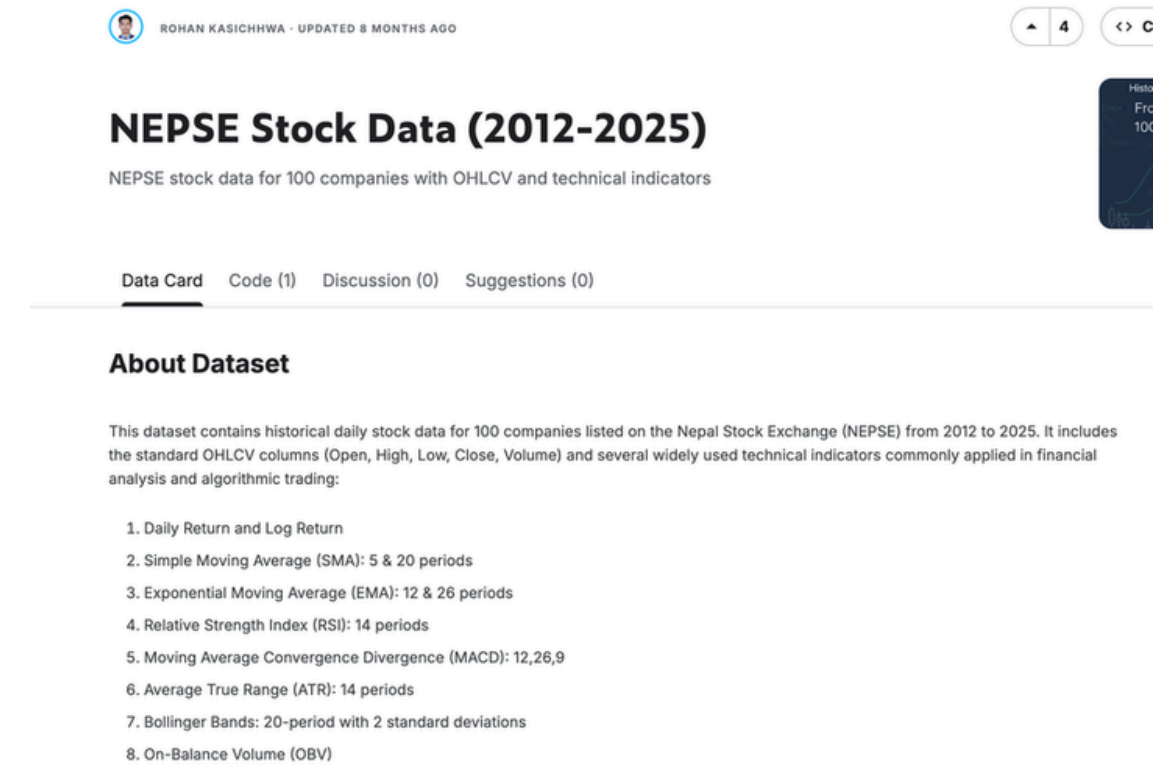
NEPSE Historical Stock Market Dataset

Variables:

- Opening Price/Closing Price
- Highest Price/Lowest Price
- Trading Volume
- Daily Price Change (%)
- Sector Type (Banking, Hydropower, Insurance, etc.)

Reason:

Suitable for predictive modeling and statistical analysis because multiple factors and historical patterns influence stock prices. The dataset allows trend analysis, visualization, and prediction of market movement.



Dataset

nepse dataset. (2022, July 10).

<https://www.kaggle.com/dinkarregmi/nepse-dataset>

NEPSE trading data. (2026, June 7).

<https://www.kaggle.com/sarojrana/nepse-data>

NEPSE Stock Data (2012–2025). (2025, September 28). <https://www.kaggle.com/rohankasichhwa/nepse-daily-stock-data-2012-2025>

Nepal Stock Exchange data. (2021, January 6). <https://www.kaggle.com/gramkrishna/nepal-stock-exchange-data>

Literature Review Summary

Stock Market Prediction Using Machine Learning Techniques (2024)

Examines machine learning models for predicting stock market trends using historical price data, trading volume, and technical indicators. The study found that historical market patterns significantly influence prediction performance.

Predicting Stock Price Movement Using Logistic Regression and Machine Learning Models (2023)

Uses classification techniques, such as Logistic Regression and other machine learning methods, to predict whether stock prices will increase or decrease based on historical trading data.

Financial Time Series Forecasting Using Historical Market Data (2025)

Focuses on forecasting market movement using historical price trends, feature engineering, and statistical analysis techniques. The study highlights the importance of trading volume and past performance in predictive modeling.

Statistical Techniques

Logistic Regression

Used to predict whether stock prices increase or decrease based on historical market variables.

ANOVA

Used to compare whether stock performance significantly differs across sectors.



Thank You